

# Experimental investigation on wireless integrated smart system for energy and water resource management in Indian smart cities

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## ABSTRACT

In the realm of India's smart cities, precise meter readings are essential for effectively managing domestic energy and water systems. Nevertheless, meter reading employing conventional techniques prove to be both costly and time-consuming, especially considering the vast user base and deficit of everyday consumption analysis. To cope with this challenge, the proposed solution introduces a novel unified wireless smart metering system for measuring energy and water usage. This prototype harnesses the smart, advanced metering technology to manage energy and water resources. The presence of data and remote control capabilities in smart distribution transformers offers distribution operators the chance to optimize system operation and control. The innovative system proposed utilizes smart metering technology and incorporates a distribution system self-healing mechanism against power outages, revolutionizing the way utilities manage water and energy consumption. The system not only offers real-time consumption statistics to the utility provider but also provides the flexibility to remotely control the system's turn-on and turn-off functions. The system effectively captures real-time data, transmits it via LoRa to Telegram application. The system automatically re-establishes remote connections in the event of disconnections caused by emergency conditions or non-payment of the bill, once the relevant issue has been resolved. Additionally, the system promptly notifies authorities on overload, over-temperature, and instances of electricity theft. The system proficiently generates bills based on the consumed data and seamlessly transmits the consumed units to the authorities. The innovative system proposed has proven its ability to communicate effectively, yielding results that align with the given circumstances.

## 1. Introduction

Urbanization is occurring at an accelerated rate worldwide, with millions of individuals annually moving from rural regions to urban centers in pursuit of enhanced living conditions and economic prospects. By 2050, as projected by the United Nations (UN), approximately 6.5 billion individuals are expected to reside in urban areas, with cities becoming home to over 60 % of the global population [1]. The International Data Corporation (IDC) forecasts a substantial growth in expenditures for smart cities on a global level. By 2022, the estimated expenses are expected to reach \$158 billion, doubling the figure of \$80 billion recorded in 2018 [2]. The rapid population growth strains existing cities and poses challenges for the future. This surge will increase demand for services and housing beyond cities' capacities, potentially causing irreparable environmental damage and worsening societal concerns. Cities grapple with resource and infrastructure strain, necessitating significant expansion to meet growing demands [3],

resulting in negative outcomes. Furthermore, issues like the COVID-19 pandemic, alongside telecom advances and electronic component proliferation, profoundly alter individuals' daily lives and perspectives across economic and social domains [4]. In response to urban expansion challenges, financial importance, and the demand for sustainability, the concept of a "smart city" has emerged. It aims to create efficient, technologically advanced, and environmentally sustainable urban environments [5].

Government and private sector smart city initiatives use Information and Communication Technology (ICT) to develop innovative solutions for urban challenges, aiming to optimize approaches for addressing complexities [6,7]. In essence, a smart city utilizes ICT and innovative technologies to boost urban management productivity and improve service standards for residents. Experts formalize the concept by examining various facets and viewpoints. One widely accepted definition describes a smart city as an interconnected system that integrates social, physical, business, and ICT infrastructure, all working together to

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elevate the city's overall intelligence and functionality [8]. Another all-encompassing definition characterizes a smart city as a progressive and forward-looking urban center landscape that effectively utilizes ICT along with diverse technologies to enhance the overall well-being and experiences of its inhabitants. Simultaneously, it focuses on bolstering competitiveness and optimizing the efficiency of urban services. Furthermore, a smart city prioritizes resource accessibility for current and future descendants, considering environmental, social, and economic factors. Leveraging these technologies and principles, it aims to foster a sustainable and thriving community [9]. Fig. 1 illustrates the characteristics of smart cities.

Early smart city initiatives aimed to improve urban residents' well-being by addressing supply and demand disparities across various areas, focusing on mitigating contradictions in service and resource availability [11]. Modern smart cities prioritize overall well-being by implementing sustainable solutions for managing water and energy resources to meet urban living demands while ensuring responsible resource management [12]. UN estimates show global water consumption rose by about 1 % annually in the 1980s and is projected to surge by up to 30 % by 2050 [13]. Urban areas consume 75 % of the world's energy [14], emphasizing urbanization's strain on water and energy resources. Predictions suggest residential and industrial sectors will drive the rising water demand [15]. Water supply efficiency is hampered by significant losses in supply systems, nearly 50 % worldwide, due to deteriorating infrastructure, inadequate billing, unreliable metering, and unauthorized usage. Addressing these issues is crucial for improving water distribution effectiveness [16] and ensuring sustainable water management practices [17]. The "water-energy nexus" offers an opportunity for integrated research, leading to innovative approaches for conserving and optimizing resources, promoting sustainability for present and future generations.

As per the World Population Prospects by the UN for 2022, it is anticipated that India's population will reach approximately 1.668 billion (166.8 crores) by the year 2050 [18]. The projections indicate that by 2050, India is expected to witness an increase of 416 million urban dwellers in its population [19]. This observation suggests that the surge in urban population is not solely attributed to overall population growth but is also influenced by migration toward urban centers. Consequently, there's a corresponding rise in energy and water consumption in urban regions. On June 19, 2019, authorities in Chennai declared "Day Zero," signifying the critical moment when the city's water distribution nearly depleted due to the drying up of all four main reservoirs [20]. Without water conservation measures in India, an additional 20 cities, including Bengaluru, Delhi, and Hyderabad, are at risk of depleting their groundwater reserves in the coming years. In

2018, India stood as the world's third-largest consumer of energy, trailing China and the United States [21], due to vibrant economic growth, population expansion, and modernization efforts. Consequently, India's energy demand has steadily risen, underscoring its crucial role in sustaining progress and development.

Water billing in India varies by region, with some areas having established systems while others use flat rates or offer subsidized supply. Similarly, electricity billing is overseen by state boards or distribution companies, handling consumption recording, bill calculation, and payment facilitation nationwide. In India, water and energy consumption are tracked through meters at consumers' premises. However, the manual reading process is labor-intensive and error-prone, leading to inefficiencies and inaccuracies in recorded data, affecting billing and resource management [22]. In densely populated countries, manual data collection becomes impractical, necessitating automated systems for monitoring usage, detecting irregularities, and providing real-time consumption statistics. Technological advancements enable efficient monitoring and evaluation of consumption data, enhancing resource management effectiveness and convenience [23].

The rising peak energy demand and the necessity to upgrade grid infrastructure emphasize the vital role of the distribution smart grid. It is essential for modernizing the aging power grid and stands to benefit greatly from smart technology integration [24]. Automating and remotely controlling the grid provides various benefits, including cost savings, enhanced data precision, and faster fault resolution within the electrical power system [25]. In the power grid, the distribution transformer is crucial for voltage transformation, converting medium voltage into lower levels used by households and commercial establishments. Distribution transformers are among the most crucial components within the distribution power system [26]. In an average-sized city with 40 high-voltage/medium-voltage (HV/MV) primary substations, there are approximately one thousand service transformers. Each year, many of these transformers suffer damage, often due to factors like elevated temperatures from reduced oil levels and overloading. Manufacturers set specific design parameters, but a component's service duration is multidimensional. For instance, briefly exceeding a thermal threshold may not cause immediate failure, but prolonged severe overloading could lead to irreparable damage. Understanding and detecting these effects in the field could help utilities reduce failure frequency and deterioration, enhancing electric service reliability and lowering operational and maintenance costs. Consequently, there is a crucial need to research and establish efficient methods for monitoring the operational conditions of distribution transformers.

Developing unified systems for Indian cities to address their growing population and urban challenges like water and energy usage is crucial. This research article focuses on introducing an integrated system for monitoring water and energy consumption, real-time monitoring of distribution transformer performance, and remote operations using IoT technology with seamless communication via LoRa.

The proposed research aims to develop and implement a system that can effectively monitor and analyze the usage patterns of water and energy. Additionally, it aims to implement strategies for monitoring the health of distribution transformers. The key contributions of the present research work are:

- Design and implementation of a cutting-edge smart water meter to monitor water consumption patterns offering remote connectivity for both connection and disconnection purposes.
- Development of a smart energy meter that provides real-time monitoring of energy usage patterns, incorporating features such as bypass detection, overload detection, with remote connection and disconnection capabilities.
- Implementation of a smart distribution transformer that continuously monitors critical parameters of oil level, temperature, and load. The distribution transformer's operation is configured to prevent

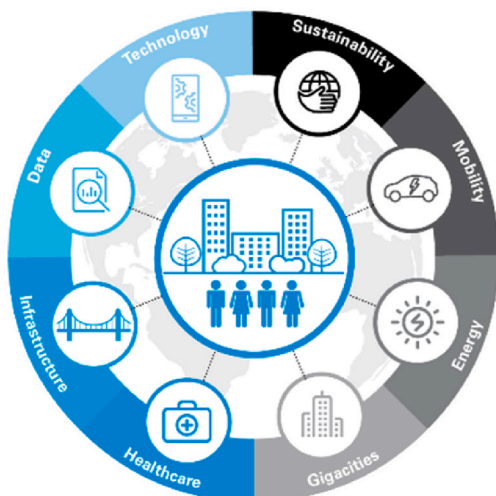


Fig. 1. Depiction of a smart city [10].

potential damage with remote connection and disconnection options.

The entire system forms an integrated system with remote operational capabilities with LoRa access. The subsequent sections of this paper discuss the following aspects:

- In-depth review of the pertinent previous research in Section 2.
- System design of proposed work in Section 3.
- Test setup, findings and discussions in Section 4.
- Further conclusions, offering suggestions for future research avenues in Section 5.

## 2. Related works

The rapid urban expansion poses challenges in managing water and energy effectively. Transitioning to smart cities emphasizes efficient resource management, crucial for urban transformation. Smart cities prioritize sustainability and efficiency, aiming to enhance residents' well-being through optimal resource use [27]. Anticipated advancements include greater autonomy and improved management of water and energy, considering local resources and stakeholder needs. IoT is essential for deploying smart city initiatives, integrating intelligence, connectivity, and data collection [28,29]. This technology makes smart cities practical by facilitating seamless communication among sensors, electronics, networks, software, and firmware. It delivers smart, dependable, and secure services. The subsequent sections offer a detailed literature review on prototypes for development.

### 2.1. Smart water meter

Water metering plays a crucial role in effectively management of water supply networks, as it accurately gauges the water distributed by the utility company to residential or business-related properties [30]. Broadly speaking, water meters can be categorized into three primary types: mechanical water meters, characterized by lower accuracy and an absence of real-time tracking capabilities; electromechanical water meters, which offer real-time data but require added protection from water; and electronic water meters, known for their high accuracy but necessitates power supply and safeguards against water ingress [31]. An electromagnetic sensor-based water flow meter was developed to gauge fluid quantity by assessing flow velocity [32]. Electrodes generate a potential difference that correlates directly with the velocity of the fluid. Regrettably, this water flow meter achieved only a modest accuracy level of 85 %. A Hall effect water flow sensing device, cloud infrastructure, Arduino Uno and Raspberry Pi was employed for a smart water meter design. The Hall effect water flow sensing device generated pulses that were received by the Arduino Uno, which then transmitted the data to the Raspberry Pi [30]. To offer users a visual depiction of the data, the raw data was analyzed by the Raspberry Pi and stored it on an internet server. This data, along with a prediction algorithm, was used for billing needs to anticipate consumer water usage needs. The use of both Arduino Uno and Raspberry Pi in the water flow meter setup added complexity to the hardware. Depending on cost and complexity considerations, either the Arduino Uno or Raspberry Pi could have been used individually. Two piezoelectric sensors were employed to differentiate between flow of cold, hot, or blended water [34]. The flow sensor exhibits an accuracy of 96 %. While the system effectively distinguishes between different water temperatures, it does not provide information about the volume of water consumed. Using just one piezoelectric transducer was adequate to perform the complete process.

A water metering system combining electrical and mechanical components, with an accuracy of 95 %, was designed for continuous monitoring of water consumption and leak detection. This system employs the General Packet Radio Service (GPRS)/Global Satellite for Mobile Communication (GSM) module [35]. However, the inclusion of a

Field Programmable Gate Array (FPGA) chip within the water flow meter to process data received from sensor adds to the total expense of the system. Ultrasonic sensor-based water meter collected data and transmitted it to the user using a GSM module, enabling real-time water flow measurements [36]. For optimal performance, this type of water meter works best in pipelines that are free of rust and debris, and to enhance accuracy, it is essential for the water conduit to be filled entirely with water. A water meter employing both laser and IR sensing device was utilized to oversee the flow velocities of coconut oil, filtered water, and glycerol [37]. The system incorporated Analog to Digital Converter (ADC) for conversion of analog sensor data to digital form, subsequently processed by an FPGA. Laser sensors have demonstrated superior accuracy when compared to IR sensors. However, it's worth noting that a water flow meter employing FPGA-based data processing tends to be relatively expensive.

A system equipped with a Hall effect sensor for measuring water flow and data processing via a Raspberry Pi 4 was employed to detect the leak [38]. Detection criteria involved identifying leakage when the measured data indicated a flow rate in liters per hour greater than zero but below a pre-established threshold for water usage. It's important to note that even in situations with low water pressure within the pipe, the system could still register water leaks. A water flow meter that is self-sufficient was developed by integrating an Arduino Uno with a Hall effect sensor, while essential operational aspects were managed remotely using a relay switch [39]. Closing the relay switch allows the system to act as an energy harvester, allowing it to charge the battery powering the entire setup. Disengaging the contactor enables the device to gauge the water consumption. It's important to note that the device can either function as a meter for measuring water usage or as a harvester of energy, this implies it cannot furnish data on total usage of water. To facilitate both flow monitoring and power generation, the water meter is equipped with water turbine generator, coupled with a Bluetooth transmitter for data transmission [40]. The Bluetooth transceiver has an effective range of approximately 10 m, making it suitable for covering a confined area. The subsequent section delves into the research contributions related to energy meters.

### 2.2. Smart energy meter

The level of energy consumption in a nation has a profound impact on its developmental quotient, symbolizing a powerful analogy for societal advancement. Discovering second possible renewable resources or implementing strategies to curb the reduction of current assets, all while preserving the environment, is crucial to address the growing demand. As previously stated, this concept can be actualized through judicious resource utilization, meticulous planning, and the pursuit of alternative technological solutions. Electricity consumption has conventionally been gauged using traditional meters employing aluminum discs. While digital meters have supplanted their analog counterparts, certain limitations endure. Traditional meters are deemed less reliable due to the necessity for consumers to allocate funds for their monthly electricity costs, coupled with the labor-intensive and costly billing process. In contrast to a conventional energy meter, smart energy meters discern the energy consumption of consumer's and provide the service supplier with supplementary data [41]. In one study [42], operational principles of traditional energy meters, and the advancement of smart energy meters with enhanced characteristics were comprehensively examined. In the context of energy monitoring, an energy meter interfaced with an AT89S52 microcontroller, retrieving and presenting the meter data on an LCD display [43]. To transmit data via SMS, a GSM modem is employed. It's important to note that the current energy meter lacks the capability to detect electricity theft. Another research study [44] investigated various communication via wireless technologies utilized in energy meters, which includes ZigBee, WiMAX, and GSM. This study recommended ZigBee as the preferred choice for smart metering infrastructure due to its cost-effectiveness and extended battery life.

Utilizing Graphical User Interface (GUI), a GPRS/GSM module and a PIC microcontroller, an Automatic Meter Reading (AMR) system was developed, enabling consumers to securely verify their identity and track their energy usage remotely from any location around the world [45]. Emphasizing enhanced energy metering and invoicing, the system prioritizes the use of an Arduino coupled with a Wi-Fi network to transmit power for each unit of data to the cloud [46], along with a GSM based smart energy meter [47]. Nonetheless, it is noteworthy that the system excludes features for detecting electrical overloads or instances of theft. Energy metres include MicroSD slots for recording and data storage [48]. Additionally, an IoT-based system has been implemented for electricity theft detection and bi-directional interaction between the meter and the cloud, facilitated by a Wi-Fi network [49]. However, it's important to note that the system lacks the capability to notify the customer in case of load limit surpassing, potentially leaving the user unaware of overloading situations. An innovative IoT-enabled smart energy meter has been designed for both household and industrial applications [50]. This innovative meter employs Arduino Uno and Wi-Fi technology, eliminating the necessity for intricate rewiring. It offers the ability to gauge power usage at the level of individual devices. The gathered consumption data is seamlessly transmitted to a ThingSpeak channel. This platform empowers consumers to conveniently access, monitor, and maintain a log of their energy consumption data. However, one notable absence in the system is the capability to detect instances of overload and potential theft.

A smart energy meter has been engineered to autonomously gauge energy consumption and seamlessly compute bills, leveraging IoT and GSM technologies [51]. Following the billing computation, an Arduino controller efficiently dispatches the invoice to the respective consumer. Concurrently, the invoice's details are also notified on the consumer's website through Wi-Fi network. A standout benefit of this smart meter is its ability to notify users when their energy usage surpasses the defined limit by sending alerts. Nevertheless, it's important to note that this system currently does not incorporate a theft detection feature. As significant as monitoring power consumption is, the following section delves into efforts directed toward minimizing power usage, focusing specifically on distribution transformer.

### 2.3. Smart distribution transformer

Transformers serve as essential components within power distribution networks, primarily tasked with transferring electric power from generating stations to the low-voltage consumers' side. As consumer energy demands escalate due to advancing technology, the lifespan of transformers undergoes a notable reduction. Failures primarily stem from overloading and overheating. The operational state of these transformers directly influences the availability of electricity across entire regions. Effectively managing the distribution grid's components becomes crucial to minimize service disruptions. To tackle these widespread issues, utility providers in numerous countries have implemented Supervisory Control and Data Acquisition (SCADA) systems [52], a computerized framework designed to monitor power, voltage and temperature. Even though, this technology minimizes the necessity for regular on-site visits by utility personnel by granting access to crucial data, it does have hindrance. Notably, vital transformer data such as load statuses or oil levels cannot be gathered. Consequently, in case of a power outage, service providers will be notified either through complaints from consumers or during routine service inspections. The introduction of technology such as IoT opens up options to handle these problems more efficiently.

A transformer monitoring system consisting of measurement units and a central master was designed to detect current, voltage,

temperature and power [53]. The measuring units communicated utilizing Radio Frequency (RF) in the 433 MHz ranges with the central unit. The GSM modem is utilized to transfer every bit of information from master unit and the measuring units to an off-site database repository. Data transfer over the internet using a GPRS connection might encounter issues stemming from the instability of both the GSM network and the service provider's internet network. A PIC microcontroller was utilized to monitor oil level, current, voltage, and oil temperature. In case of any anomalies detected, the system transmitted this information via Short Message Service (SMS) using GSM technology to relay data between the sensors and the central control room [54]. While the system notifies the control board of abnormalities through SMS, it does not possess the remote connection/disconnection feature to prevent further damage to the transformer. This is crucial, especially considering that the service provider may require a certain amount of time to reach the transformer's location. The expansion of distribution transformer monitoring was made possible by integrating IoT technology, along with the incorporation of vibration sensors, oil sensors, and humidity sensors [55]. A PIC controller facilitated interaction with the sensors, while a GPRS network transmitted the gathered data to a web interface accessible to utility providers. Notably, this system was activated solely during emergencies and lacked support for monitoring the transformer in real-time.

A transformer parameter monitoring system utilizing IoT was developed to oversee voltage, temperature, and current [56]. The system utilizes a PIC microcontroller and Wi-Fi technology to gather consumption data, transmitting it effortlessly to a ThingSpeak channel. A comparable system was devised employing Arduino, boasting added functionalities such as a sound alert and a relay mechanism to disconnect specific loads in the event of overloading [57]. Nevertheless, the system lacks an alert feature for abnormalities, requiring continuous monitoring of the ThingSpeak channel to detect any irregularities. Thermal monitoring and safeguard system based on IoT was suggested, involving a WiFi-enabled microcontroller to connect humidity, temperature, and current sensors [58]. Additionally, a direct current fan was incorporated to cool down overheating situations. To prevent complete shutdown and load shedding, tripping mechanism using a three-phase logic was introduced. Although cost-effective, this system encountered delays during the testing phase. A proposed remote transformer system employing an ARM Mini2440 was designed to supervise critical parameters like oil level, temperature, and overloading of the transformer. Utilizing a public network, the system communicates observed data, accessible for analysis via the internet [59]. Moreover, a microphone was integrated to capture the transformer's humming noise. However, the system currently lacks an emergency alerting feature. Despite the abundance of literature on real-time transformer monitoring, the automatic detection of faults and anomalies remains a persistent challenge. Integrating water and energy meters enables users to conveniently monitor both water and energy usage. The following section examines the relevant research conducted on the integrated water and energy meter.

### 2.4. Integrated system

Energy and water stand as the most prevalent resources, yet when unmonitored and uncontrolled, they pose significant waste potential. It's pivotal to measure and manage consumption of these resources as it yields multifaceted benefits. While energy and water meters have received considerable attention, integrated systems for both resources have seen comparatively lesser focus. Selecting the most appropriate communication network is also crucial. The subsequent section delves into the LPWAN network.

## 2.5. Communication protocol

The cornerstone of IoT systems revolves around the area of coverage facilitated by the wireless communication network utilized, specifically distinguishing between short-range and wide-range networks. Beyond mere range considerations, when factoring in power consumption and energy efficiency, the main classifications recognized are Low-Power Short-Area Networks (LPSAN) and LPWAN [60]. LPSAN are tailored for cost-effective, short-range coverage with energy efficiency, specifically addressing the challenge of establishing wireless connectivity within the “last 100 m,” which involves linking neighbourhood connected devices to the world of Internet via a more organized Wide-Area Network (WAN). Typically, access to WAN is facilitated through strategically positioned and specialized gateways [61]. LPWAN systems, a more recent development compared to LPSANs, hold significant importance in providing distinct IoT services commonly termed as massive Machine Type Communications (mMTC) [62]. In contrast to LPSAN, LPWAN aim for extensive coverage spanning several kilometres, achieved with low-cost and low-power consumption methods [63].

Long Range Wide Area Network (LoRaWAN) emerges as an LPWAN technology particularly well-matched for smart meters [64]. Long Range (LoRa) networks, like those incorporating the LoRaWAN standards, bridge the technical disparity failed to be included by cellular, Wi-Fi, and Bluetooth Low Energy (BLE) networks. Unlike these wireless networks that use up more bandwidth or power, possess limited range, or struggle with indoor penetration at significant depths, LoRa technology stands out. LoRa devices, in conjunction with the LoRaWAN standard, excel in diverse settings, indoors and outdoors, encompassing both smart city initiatives and metering applications. While 5G boasts superior speed and connectivity, LoRa network shine with a communication range surpassing 25 km, establishing themselves as an adaptable, reliable, and cost-effective choice for both outdoor and indoor uses. Blending the advantages of cellular and Wi-Fi technologies, LoRa provides indoor connectivity within unlicensed bands and ensures end-to-end encryption from consumer devices to application servers for out-of-doors applications [65]. The proposed system selects the LoRa-based network due to its proven efficacy in utilizing sensors, information in real-time, user interfaces, software applications, and communication infrastructure.

Examination of the literature highlights several challenges faced by existing water meters, including significant upfront investment costs, absence of remote connectivity, and subpar precision. Likewise, the energy meters within smart city frameworks encounter constraints in identifying electricity theft or instances of overloads. Furthermore, the Smart Distribution Transformer systems either lacked real-time monitoring or have failed to provide alerts during emergencies. Moreover, there exists a necessity to optimize wireless protocols to expand the coverage area of service.

Above all, there's a need for an integrated system to streamline the entire process, viz., real time monitoring of water and energy usage, remote connection and disconnection of systems, power theft identification, temperature and overload conditions of distribution transformer. The current research work focuses to bridge these gaps by developing an innovative integrated system capable of remotely managing connection status, overseeing water and energy usage, monitoring transformer parameters, and identifying critical conditions like overloading, electricity theft, and temperature fluctuations in transformers. The goal is to minimize costs for both customers and regulatory bodies. The present work aims to develop an integrated system encompassing:

- Smart water meter employing a precise water flow meter to precisely gauge water usage, enabling the measurement of water usage in total and the generation of corresponding water bills based on consumption.
- Smart energy meter that computes electricity usage by assessing various parameters like voltage, frequency, power, power factor, current, and energy. It facilitates billing based on the energy consumed while also identifying voltage overloads to avert penalties and detecting instances of theft of electricity by users.
- A smart distribution transformer designed to oversee critical parameters viz., voltage, current, power, and temperature. In the event of temperature spikes or overloading, it can sever the connection between the load and the transformer to prevent further issues.

Moreover, the present work endeavours to establish connectivity between the smart water meter, smart energy meter, and smart distribution transformer through communication enabled by LoRa technology. The uniqueness of the integrated system will be elaborated in the following section.

## 2.6. Novelty of the integrated system

Based on the literature survey, it is noted that only a limited number of contributions utilize LoRa communication [71–73], offering coverage spanning 5–8 km, surpassing Bluetooth, Wi-Fi, and Zigbee. Moreover, LoRa operates without licensing requirements, boasts low power consumption, and presents a cost-effective solution. “The smart water meter is ingeniously designed in this work to employ LoRa for seamless real-time data transmission between itself and the smart energy meter, spanning a distance of up to 5 km. This surpasses the capabilities of smart water meters utilizing Bluetooth, Wi-Fi, and Zigbee for communication making it a novel system”. Very few works reported in literature [38&40] discuss the adjustment of calibration factors for water flow sensors but do not specify the exact values used in their studies. “The calibration factor in the smart water meter design is adjusted to ensure the meter delivers accurate readings”. Literature [46–50] highlights a deficiency in existing smart energy meters, namely their inability to detect overloading situations. “The designed smart energy meter in this work, a mechanism for detecting overload situation is also implemented, thereby enhancing the performance of the smart energy meter”. The literature on smart distribution transformers reveals shortcomings such as the absence of remote connection/disconnection features [53,54,56] or real-time monitoring [55]. “In contrast, the proposed smart distribution transformer encompasses remote connection/disconnection features and real-time monitoring of parameters such as voltage, current, power factor and temperature. The present contribution is a novel contribution with respect to smart distribution transformer presented in this work. The key highlight lies in the seamless integration of smart water meter, smart energy meter and smart distribution transformer, thus making it a novel system. Earlier investigators have not explored this part of the work. The integration of all these modules has been attempted in this work, making this work a novel contribution. It not only combines these components, but also ensures the incorporation of essential features like remote connection/disconnection, real-time monitoring, theft detection, overload detection and communication with consumer as well as service provider”. In the subsequent section, the block diagram of the comprehensive experimental setup, along with the individual modules and their functionalities, is elaborated.

### 3. System design and methodology

In contemporary times, establishing a wireless network across extensive areas poses minimal challenges, owing to technological progress. The development of viable systems for measuring water, energy, and monitoring transformers in smart cities results from the fusion of wireless components with compatible IoT controllers. The block diagram of the present work and methodologies for an integrated system that tracks household water and energy usage, along with monitoring transformers are discussed in the subsequent sections.

#### 3.1. System design

The experiment setup devised in this study aims to develop smart meters for both water and energy, with a focus on monitoring consumers utilization while factoring for the sensor node's energy consumption. Additionally, it aims to develop a smart distribution transformer for overseeing the transformer's health. Fig. 2 illustrates the proposed integrated system's structure and Fig. 3 presents the general architecture

of the integrated system. The project involves a case study focusing on two household scenarios. The entire system is developed taking into account various parameters related to water and energy usage in both houses along with monitoring the distribution transformer.

A residential setup includes both a smart water meter, responsible for monitoring water consumption, and a smart energy meter, which tracks energy usage and interfaces with the smart water meter. In addition to measuring the energy usage, the smart energy meter also retrieves information from the smart water meter, facilitating seamless transmission of combined information to the vendor upon request. The smart distribution transformer is linked to numerous loads and constantly monitors its health. All gathered data is transmitted to the utility center. In case of overload or a temperature increase, the transformer is disconnected. Each unit establishes connections with both the service provider and other units through the LoRa network. To link the LoRa gateway and the network server, the Message Queuing Telemetry Transport (MQTT) protocol is utilized. A device employing LoRa technology transmits frames and packets in a multi-directional manner when communicating via the internet. These packets are then received by one

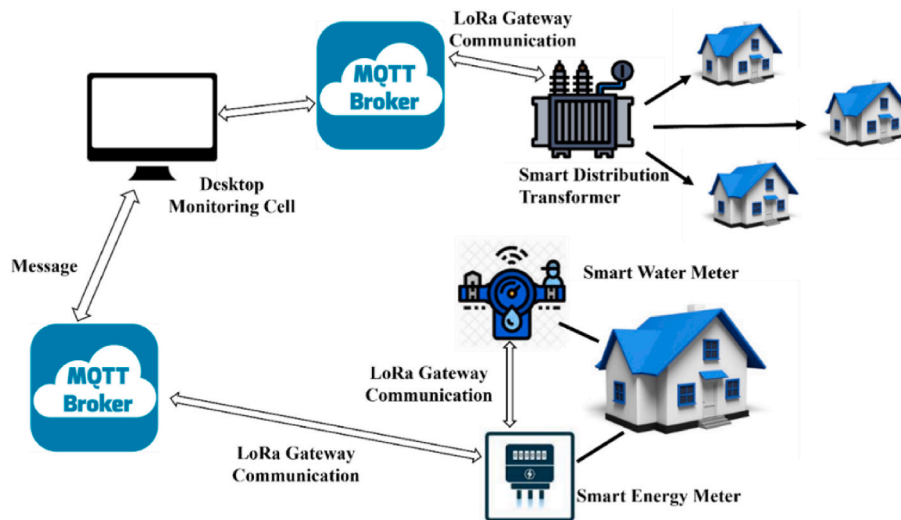


Fig. 2. Structure of the integrated system.

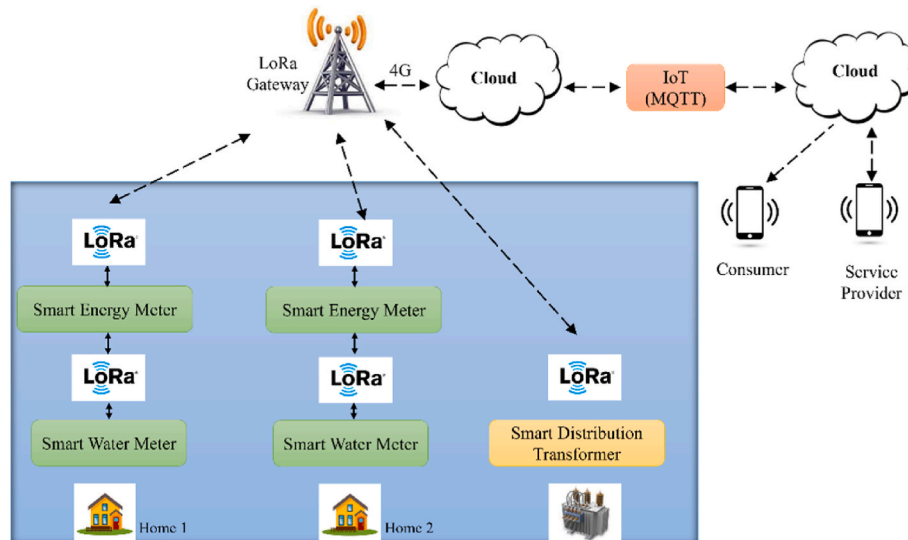
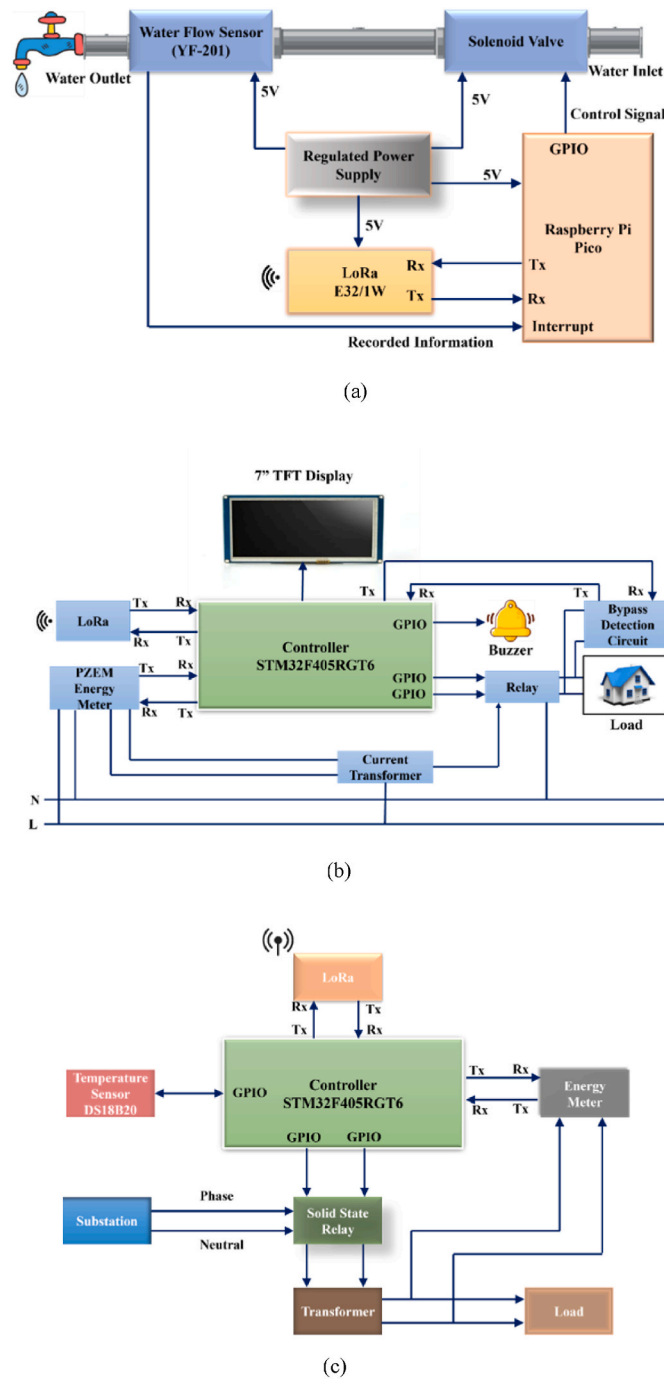


Fig. 3. Architecture of the integrated system.



**Fig. 4.** Block diagram depicting distinct components of the integrated system (a) the smart water meter, (b) the smart energy meter and (c) the smart distribution transformer.

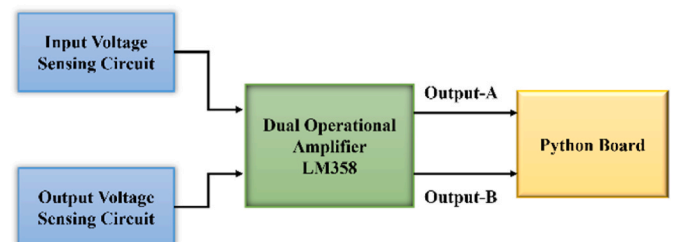
or more Gateways. Gateways demodulate the LoRa packets before relaying them to the network server via the internet. In numerous LoRa setups, the MQTT protocol facilitates communication between the network server and the Gateways. Messages between devices are transmitted utilizing the publish-subscribe protocol of MQTT. The MQTT protocol comprises the “MQTT broker” server, responsible for gathering messages, and clients that possess the capability to both read from and write to this broker. As part of the publish-subscribe protocol, clients are required to specify the topic that they intend to communicate about and subscribe to. Employing the MQTT protocol offers the benefit

of reduced bandwidth requirements due to its binary format, while its compact implementation minimizes power consumption [66].

Fig. 4 displays the schematic diagram of the distinct systems within the integrated setup. In the smart water meter as seen in Fig. 3(a), YF-201 Hall effect water flow sensor, a Raspberry Pi Pico, and a LoRa module- E32/1W are employed. Conversely, as illustrated in Fig. 3(b), the smart energy meter integrates a STM32 controller, a PZEM-004T energy meter, and a bypass detection circuit. As for the smart distribution transformer, as shown in Fig. 3(c), it incorporates a temperature sensor for monitoring transformer temperature, alongside a STM32 controller, LoRa module, and PZEM-004T energy meter for monitoring parameters such as current, voltage and frequency. The core of the system, comprising a STM32 controller and Raspberry Pi Pico, performs all essential functions. Real-time data collection involves linking a YF-201 sensor directly to the Pico controller. Additionally, the STM32 controller retrieves data on energy usage from the PZEM energy meter and the temperature sensor. The control of the solenoid valve within smart water meters and the relay operations in both the smart energy meter and smart distribution transformer is directed by the Pico controller and STM32 controller, respectively. Meanwhile, the Raspberry Pi Pico oversees the data capture from the YF-201 sensor, tracking water consumption, while the STM32 controller handles and archives energy readings. Upon request or upon detection of an interrupt by the sensor, the control board receives data via LoRa module, prompting the LoRa to awaken from sleep mode for remote data access. The PZEM-004T facilitates the measurement of parameters like power, voltage, power factor, frequency, and current by real-time monitoring of AC main lines. It offers two options for current measurement: a 10 A range with inbuilt shunt resistor and a 100 A range with external current transformer. For the 100 A range module to function, the neutral line from the load must traverse through the current transformer's core. Fundamentally, the PZEM-004T is driven by the V9881D, a single-phase energy meter with a system-on-chip design. Voltage measuring range spans from 80 to 260 V with a resolution of 0.1 V, boasting a measurement accuracy of 0.5 % [67]. The subsequent section delineates on the functionality of the Bypass detection circuit.

### 3.2. Bypass/tamper detection circuit

The supplied voltage before and after the PZEM energy meter in a smart energy metering system are captured by the input and the output voltage detecting circuit respectively. Fig. 5 displays the schematic visualisation of the bypass/tamper detection circuit, comprising a sensing circuit composed of a rectifier, optocoupler, voltage divider, and capacitor. The output from the sensing circuit is directed to a dual operational amplifier. This component assesses both the outputs and generates a logical output contingent upon the disparities between the parameters. The controller receives the logical outputs to identify anomalies, such as potential electricity theft. The logical outputs and the corresponding actions taken by the controller in response are outlined in Table 1.



**Fig. 5.** Schematic diagram depicting the bypass/tamper detection circuit employed in smart energy meter.

**Table 1**  
Controller's response to different voltage conditions.

| Output of dual operational amplifier |          | Remarks           |
|--------------------------------------|----------|-------------------|
| Output-A                             | Output-B |                   |
| 0                                    | 0        | No power          |
| 0                                    | 1        | Theft detected    |
| 1                                    | 0        | Line Disconnected |
| 1                                    | 1        | Line Connected    |

Combining various components creates a unified system comprising a smart water flow meter, smart energy meter, and smart distribution transformer. Once integrated within a single enclosure, these components are interconnected to measure and monitor both energy and water consumption. The next section elucidates the sequence of actions, detailing the functioning of each module.

### 3.3. Working of the experimental setup

The transmission of data from the water meter to the energy meter via LoRa. Subsequently, the energy meter communicates both water and energy data to the utility centre. Likewise, the parameters of the distribution transformer are shared with the utility centre. Each module's operation is elaborated upon in detail in the subsequent sections.

#### 3.3.1. Smart water meter

The water flow sensor registers motion within the water conduit, while the Hall effect within the water flow sensor translates this motion into pulses. When the pipeline valve is shut, indicating no water flow, a zero output is identified as there are no pulses being generated. The

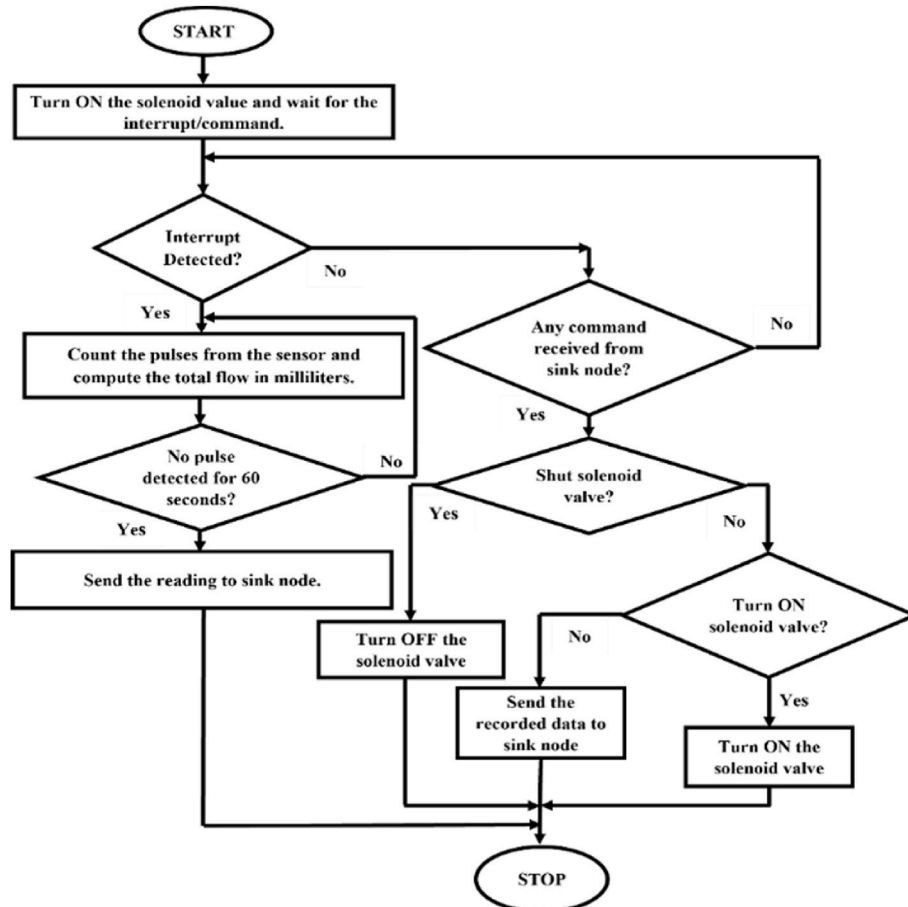
sensor translates the Hall effect motion caused by flow of water while the valve is open into pulses, subsequently converting these pulses into a measurable value. The calibration factor mirrors the Hall-effect flow sensors output and is adjustable to enhance accuracy. Alterations in pressure of water and diameter of the pipe prompt modification in this coefficient. As there's no assurance of a cycle's completion precisely within a second, hence the resulting output is quantified in milliseconds (ms) from the previous execution. The flow rate calculation considers the frequency of pulses/second for each unit of measurement, alongside the calibration coefficient influencing the rate of flow as described in equation (1).

$$\text{Flow rate} = \frac{\left( \left( \frac{1000}{\text{Current time} - \text{Previous time}} \right) \times \text{Current pulse count} \right)}{\text{Calibration factor}} \quad (1)$$

Determining the current measured water flow in milliliters (ml) becomes simpler once the flow rate is established, as outlined in equation (2).

**Table 2**  
Adjustment of calibration factor.

| Iterations | Calibration Coefficient | True Value (ml) | Measured Value (ml) |
|------------|-------------------------|-----------------|---------------------|
| 1          | 3                       | 2000            | 4050                |
| 2          | 3.5                     | 2000            | 2800                |
| 3          | 4                       | 2000            | 2250                |
| 4          | 4.5                     | 2000            | 2010                |
| 5          | 5                       | 2000            | 1800                |
| 6          | 6                       | 2000            | 1000                |



**Fig. 6.** Flowchart presenting the various stages of the proposed process.

$$\text{Measured flow} = \left( \frac{\text{Flow rate}}{60} \right) \times 1000 \quad (2)$$

Therefore, the total water flow comprises the sum of the current reading and the previous readings, measured in milliliters. All the parameters used in determining the total water consumption are initialized by the controller to “0” to ensure no impact on billing in a newly installed meter. The approach used follows a series of steps outlined in Fig. 6, as follows:

- The solenoid valve is triggered by the driving circuit to open upon the reception of a 10 ms pulse from the Pico controller.
- The Pico controller receives the output of water flow sensor through its interrupt pin, tallying the interrupts to determine the pulse count per unit of time. The water flow rate corresponds directly to the number of interrupts, allowing the calculation of the total water flow in ml using the pulse count acquired from the sensor.
- The total measured volume will be stored for further calculations if no water flows through the solenoid valve for a predefined duration, like 60 s.
- The solenoid valve is remotely deactivated in case of delayed or non-payment of the bill, and it is activated again once the bill is successfully paid. This remote control enables the manipulation of the solenoid without involving physically visiting the site.
- The power consumption is minimized by reducing the unit’s operational hours. Consequently, when no operations are active, the LoRa module enters sleep mode.

Improving accuracy involves a trial-and-error method to select the appropriate calibration coefficient value. Accuracy increases as the vicinity between actual and measured values improves. Table 2 displays the measured and true values corresponding to various calibration

coefficients.

The calibration factor of 4.5 performs optimally when the actual and measured values are close. Proper adjustment of the calibration factor is meant to ensure a minimum accuracy level of 95 % for the system. Through a trial-and-error approach, 2000 ml of water is run via a sensor, and the observed value is compared to the real value after initially setting the calibration coefficient to 1. The process is then iterated, incrementally adjusting the calibration coefficient until the measured value aligns as closely as achievable with the actual value. The present research achieves a heightened level of accuracy for the calibration coefficient, reaching 4.5 through a trial-and-error method. The sensor requires a water pressure of 0.5 bar for optimal functioning as a preventive measure against the sensor generating pulses caused by air flow inside the water pipe. All parameter values are refreshed using the latest measurement, establishing a foundation for measurements that follow.

### 3.3.2. Smart energy meter

The STM32 controller serves as smart energy meters brain, overseeing all its functionalities. It processes data that is then showcased on the Liquid Crystal Display (LCD) and communicated via a LoRa gateway to the service provider. A Solid-State Relay (SSR) is utilized either to “Disconnect” or “Connect” the load each time the smart energy meter receives commands from the electricity board. Detection of power theft at the consumer side is accomplished through the bypass/tamper detection circuit when the consumer tries to bypass the meter. The flow of operations executed in the smart energy meter is depicted in Fig. 7.

The operational process of smart energy meter unfolds as follows:

- Continuous monitoring of energy levels will be showcased on the Thin Film Transistor (TFT) LCD display. This display will also exhibit data regarding water consumption received from the smart water meter, accompanied by alert messages.

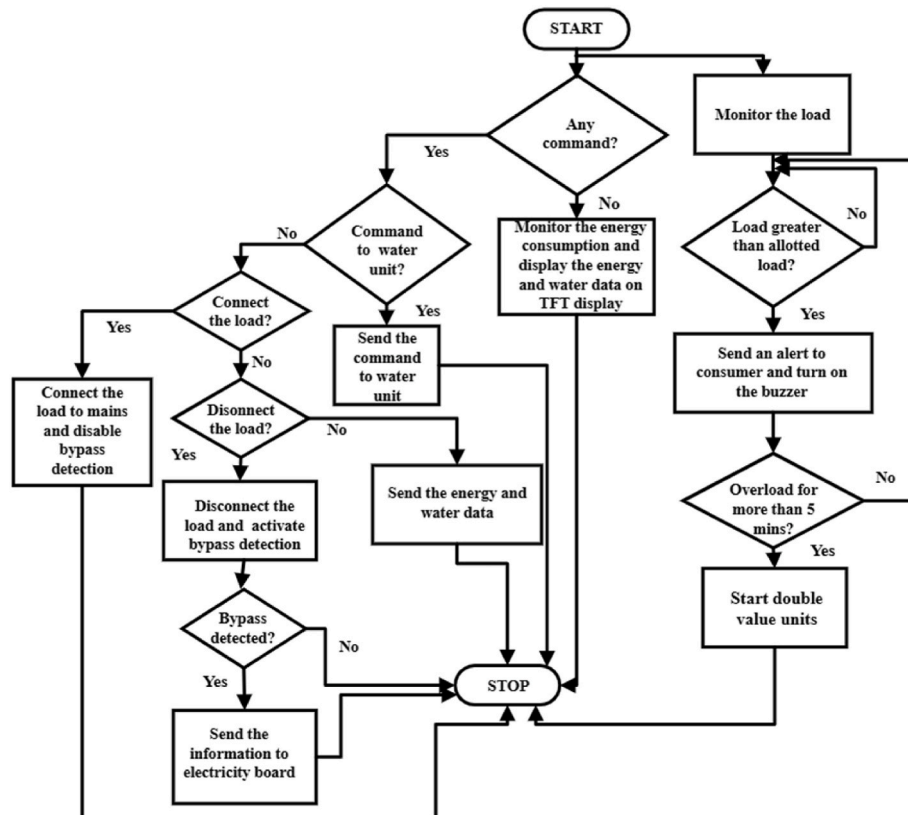


Fig. 7. Flowchart delineating the sequential stages within the process.

- When prompted by the water regulatory board, communication is initiated by smart energy meter with the smart water meter. This facilitates tasks like activating or deactivating the solenoid valve and retrieving water consumption data. Subsequently, the water regulatory board receives the data forwarded by the smart energy meter.
- When the consumed load surpasses the designated limit, the user receives both a buzzer notification and a message. If this overload condition endures for over 5 min, the price/unit will be doubled.
- Upon unsuccessful bill payment, the smart energy meter receives instruction from the electricity board for disconnection of the line and activate tamper detection. If the bypass/tamper detection circuit identifies any tampering, an alert is delivered to the electricity board by the smart energy meter. Upon receiving the directive from the electricity board regarding successful bill payment, the line is connected to the load remotely.

The smart energy meter is combined with the smart water meter to facilitate the transmission of data on both energy and water consumption, as well as billing information.

### 3.3.3. Smart distribution transformer

The energy meter gathers real-time voltage and current data, forwarding it to the STM32 controller. Additionally, the STM32 controller collects information from the temperature sensor. When commands arrive smart distribution transformer from the electricity board, the SSR

comes into play, facilitating either the “Connect” or “Disconnect” functions for the load. Moreover, in instances of rise in temperature or overload, the solid-state relay plays a role in disconnecting the transformer from the substation. Fig. 8 illustrates the method of operation employed for the smart distribution transformer, outlining the actions carried out in the following manner:

- Three concurrent operations take place: monitoring parameters, tracking temperature, and awaiting commands from the electricity board.
- The board’s commands entail either remotely activating or deactivating the transformer, or transmitting the various parameters of the smart distribution transformer.
- In the event of load drawing excessive power beyond the allocated limit, the transformer is promptly disconnected from the substation, and an alert is dispatched to the board to report this occurrence. Following this, the transformer can be reactivated after a specific duration, or alternatively, a repair person may be dispatched to the site for further investigation.
- The transformer’s temperature undergoes constant monitoring. If an increase surpassing the threshold temperature is detected, the transformer is disconnected from the substation, and an alert is dispatched to the board to initiate further actions.

The control board utilizes the following command format to send a read command to retrieve sensed data: slave’s address + 0 × 04 +

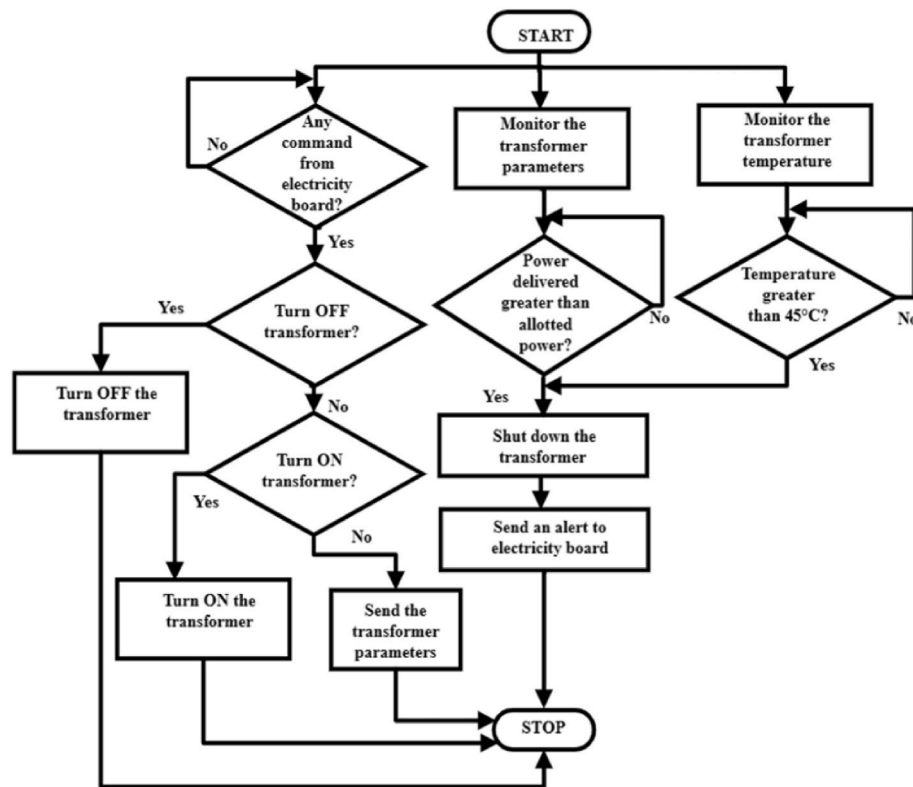


Fig. 8. Flowchart depicting the sequential stages of the process.

register address's higher byte + register address's lower byte + # registers-higher byte + # registers-lower byte + Check Redundancy Code (CRC) -higher byte + CRC -lower byte. In a single slave scenario, the address of the energy meter acts as the address of the slave, with 0xF8 designated as the general address. After specifying the register address to be read, the instruction for reading the input register follows denoted by  $0 \times 04$ . The term "# registers," paired with the CRC, signifies the count of registers slated for reading. Voltage and current values reside within the first and third registers of the 16-bit initial register. Subsequently, the fourth and fifth registers hold power values, the sixth and seventh registers store energy values, the eighth register contains frequency data, and the ninth register stores power factor information. These values are maintained in hexadecimal format, and post shifting and logic OR operations, they undergo conversion to decimal for display. The subsequent section details additional findings resulting from the conducted experimental research.

#### 4. Experimental setup, results and discussions

The following section details the experimental setup along with results and discussions. The evaluation of the system's effectiveness is conducted via testing the system for efficient communication and accuracy of the parameters.

##### 4.1. Smart water meter

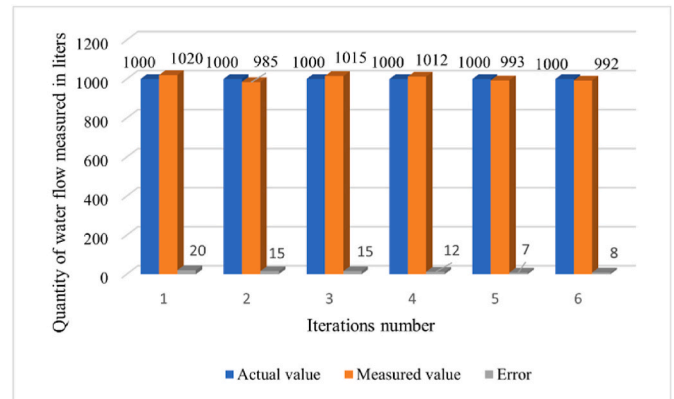
To evaluate the precision of a water flow meter in quantifying the water's volume passing within a pipeline, the experimental setup is linked to the inlet line of the household water supply. Fig. 9 illustrates the hardware configuration. The system undergoes testing by circulating water through a 1/2-inch conduit. The smart energy meter obtains the water flow measurement value and showcases it through the utilization of a LoRa network. Addressing the power consumption and management for sustained operation poses a challenge for wireless sensor networks, especially in the context of smart water meters. Relying on externally powered devices is not advisable from the utility provider's standpoint. Additionally, the placement of water meters is frequently distant from readily available sources of electricity. Solar energy stands out as one of the widely recognized and environmentally sustainable renewable energy sources for energizing embedded systems and wireless sensors. The

**Table 3**  
Adjustment of calibration factor.

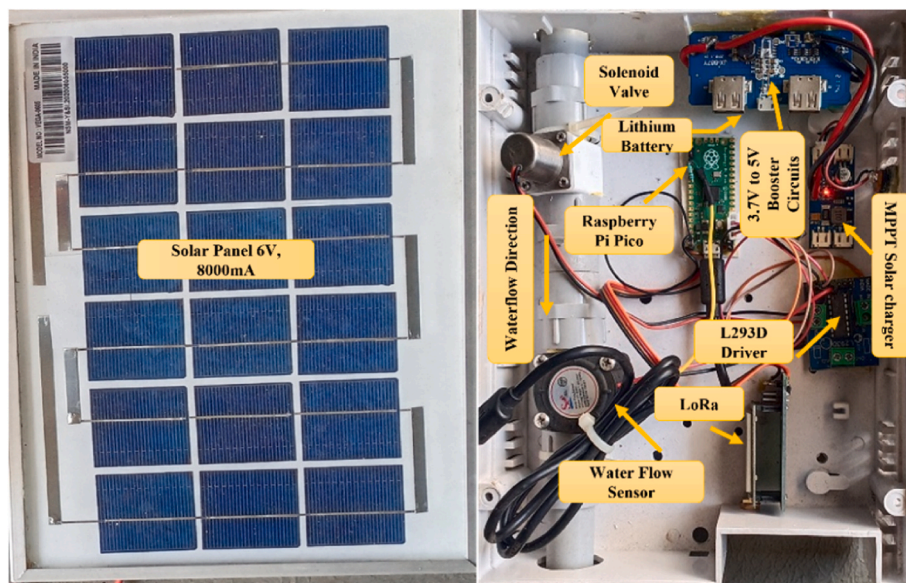
| Iteration Count | True Value | Measured Value | Accuracy |
|-----------------|------------|----------------|----------|
| 1               | 1000       | 1020           | 98.03 %  |
| 2               | 1000       | 985            | 98.5 %   |
| 3               | 1000       | 1015           | 98.52 %  |
| 4               | 1000       | 1012           | 98.81 %  |
| 5               | 1000       | 993            | 99.3 %   |
| 6               | 1000       | 992            | 99.2 %   |

optimal choice that powers up the embedded components in wireless sensor networks is a solar cell, thanks to its remarkable power density.

Multiple iterations involve conducting tests with a quantity of 1000 ml of water to evaluate the accuracy of the water flow meter. As indicated in Table 3, a relative accuracy of 98 % is recorded by the smart water meter. Fig. 10 illustrates that, across all five iterations, the measured values closely align with the actual value. The experimental setup for the smart water meter is tested for two house scenarios.



**Fig. 10.** A comparison between the actual volume of water flow and its measurement, employing a calibration factor of 4.5.



**Fig. 9.** Smart water meter prototype.

#### 4.2. Smart energy meter

The smart energy meter is configured to read the consumed energy of designated premises and communicate this data to the service provider. Moreover, the smart energy meter is designed to figure out incidents of consumer-side power theft and has the capability to remotely establish or sever the connection between the load and the line. The experimental setup of the same is illustrated in Fig. 11.

Various parameters such as voltage, current, power, power factor, frequency, and energy are displayed on the TFT LCD screen. To enable the “Connection” or “Disconnection” between the line and the load, the SSR is employed. To detect electricity theft in the consumer load, the smart energy meter utilizes bypass/tamper detection circuit. In the event that the consumer establishes a direct connection between load and line, bypassing the meter, the bypass/tamper detection circuit sends a digital high signal to the controller. Subsequently, the controller relays an alert to the service provider. The initial design of the smart energy meter experimental setup was tailored for a single-home scenario and later expanded to a two-home scenario upon achieving the anticipated results. The experimental setup underwent testing using two 75W bulbs each. In the case of an overload condition, the load was connected with

two 100W bulbs.

#### 4.3. Smart distribution transformer

The smart distribution transformer incorporates a STM32 controller-enabled control board that actively monitors the transformer’s voltage, current, and temperature. In the event that either the power or temperature surpasses predefined threshold levels of 300W or 39 °C, the transformer is automatically disconnected from the substation to prevent additional damage. Simultaneously, an alert message is sent to the control center, providing real-time information on the transformer’s status. Upon the values dropping below the established thresholds, the transformer can be remotely reactivated. This eliminates the need for manual intervention from the control board, streamlining the management and maintenance processes. The smart distribution transformer underwent testing with a load equivalent to two 100W bulbs. To test for overloading conditions, three 100W bulbs were utilized. A visual representation of the hardware prototype of the smart distribution transformer is shown in Fig. 12.

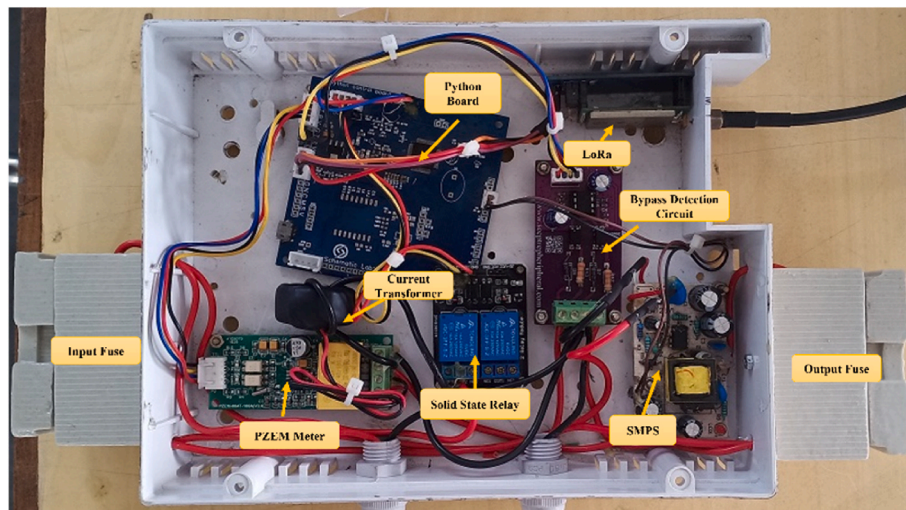


Fig. 11. Smart energy meter prototype.

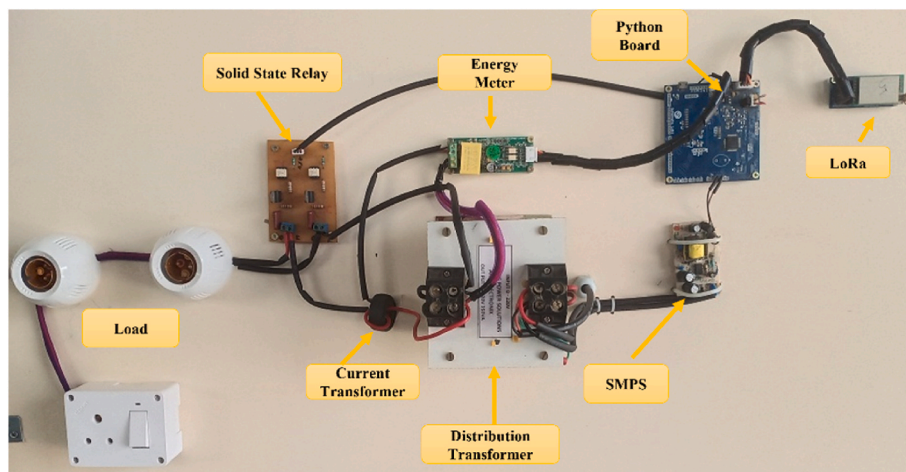
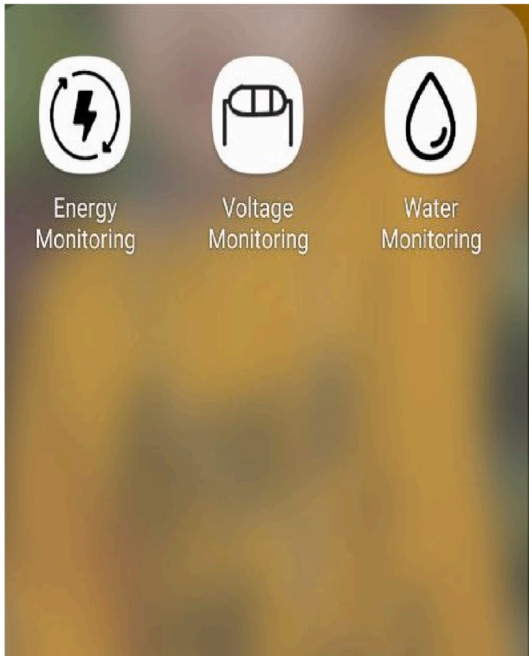


Fig. 12. Smart distribution transformer hardware module.

4.4. Data gathering

Three Android applications have been developed to facilitate the subscription and publication of MQTT messages. These apps serve the purpose of presenting real-time information on water data, energy data, and transformer data. The sender of messages is referred to as the

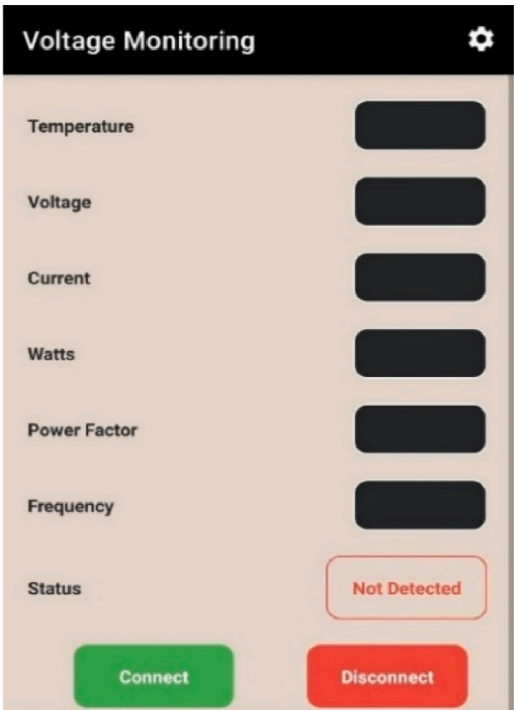
publisher, while the recipient is known as the subscriber in this communication model. In the publish/subscribe model, the customer sending a message operates independently of the clients that receive the message. Direct communication between publishers and subscribers is not feasible. In fact, both the publisher and the subscriber are unaware of each other's existence. A third-party entity, known as the broker, is



(a)



(b)



(c)



(d)

Fig. 13. MQTT applications: (a) Three applications for displaying energy, transformer and water data respectively, (b) Splash screen of energy monitoring app, (c) Splash screen of transformer parameter monitoring app and (d) Splash screen of water monitoring app.

The figure displays three screenshots of MQTT application settings forms, labeled (a), (b), and (c). Each form has a title bar with a gear icon and a 'Settings' title. The forms are organized into sections with labels and input fields.

**(a) Energy Monitoring**

- MQTT Broker Address: tcp://test.mosquitto.org
- MQTT Port Address: 1883
- MQTT Client Name: ENERGY APP
- Telegram API Token: 6347204760:AAEg3tG0a5DJvg3wTNI9kTMH5he0Q-4sIL8
- Telegram Bot Id: (empty)
- Telegram User1 Chat Id: -920786643
- Telegram User2 Chat Id: -914098956
- Board Telegram Chat Id: (empty)
- Home Publish Topic: all\_recv
- Home-1 Subscriber Topic: Home1\_ele\_status
- Home-2 Subscriber Topic: Home2\_ele\_status
- User Name: (empty)
- Password: (empty)
- Buttons: Save, Clear

**(b) Settings**

- MQTT Broker Address: tcp://test.mosquitto.org
- MQTT Port Address: 1883
- Telegram API Token: 6347204760:AAEg3tG0a5DJvg3wTNI9kTMH5he0Q-4sIL8
- Telegram Bot Id: (empty)
- Telegram User1 Chat Id: -920786643
- Telegram User2 Chat Id: -914098956
- Board Telegram Chat Id: (empty)
- Home Publish Id: all\_recv
- Home-1 Subscriber Id: Dist\_transformer
- User Name: (empty)
- Password: (empty)
- Buttons: Save, Clear

**(c) Settings**

- MQTT Broker Address: tcp://test.mosquitto.org
- MQTT Port Address: 1883
- Telegram API Token: 6347204760:AAEg3tG0a5DJvg3wTNI9kTMH5he0Q-4sIL8
- Telegram Bot Id: (empty)
- Telegram User1 Chat Id: -920786643
- Telegram User2 Chat Id: -914098956
- Board Telegram Chat Id: (empty)
- Home Publish Id: all\_recv
- Home-1 Subscriber Id: Home1\_wtr\_status
- Home-2 Subscriber Id: Home2\_wtr\_status
- User Name: (empty)
- Password: (empty)
- Buttons: Save, Clear
- Version: v1.4\_Release

Fig. 14. MQTT applications: (a) Subscription of energy monitoring app, (b) Subscription of transformer parameter monitoring app and (c) Subscription of water monitoring app.

tasked with overseeing the communication between the subscriber and the publisher. To ensure their proper distribution to subscribers, the broker filters all incoming messages. The applications and its splash screen are shown in Fig. 13. Regulatory boards oversee the control of applications, granting them the ability to remotely connect, disconnect, and capture readings for sending billing information. The applications also display real-time data that is captured.

The settings related to the subscription of user are shown in Fig. 14. The MQTT protocol offers a lightweight approach to facilitate messaging through a publish/subscribe model, making it well-suited for IoT communication. It is particularly effective for scenarios involving low-powered sensors, mobile devices like cellphones, microcontrollers, and embedded computers. An MQTT broker serves as an intermediary entity facilitating communication among MQTT clients. More precisely, it receives messages published by clients, filters these messages based on topics, and subsequently distributes them to the respective subscribers. Eclipse Mosquitto is a message broker that operates on an open-source model. An MQTT broker plays a crucial role by overseeing essential functions such as message routing, ensuring scalability, maintaining security measures, facilitating seamless integration, and managing session interactions. Ports serve as digital communication endpoints essential for transmitting and receiving data across networks. MQTT ports specifically enable communication between MQTT clients and servers, acting as channels through which MQTT messages traverse. Each MQTT port is associated with a distinct service, and multiple ports can be operational concurrently.

The port 1883 is designated for unencrypted MQTT connections,

serving as the predominant and default port for most MQTT brokers. This widely used port enables MQTT clients to publish messages, subscribe to topics, and receive published messages, facilitating essential communication within the MQTT protocol. Application Programming Interface (API) are systems that facilitate communication between two software components by employing a predefined set of definitions and protocols. This API enables the creation of personalized Telegram clients, enabling communication of collected data from the MQTT application with users. Topics are structured strings that categorize the subject of a message in a hierarchical manner. Publishers, when sending messages to the broker, assign them to a particular topic. Subscribers indicate their interest in receiving messages by subscribing to one or more MQTT topics. The broker subsequently directs messages to the relevant subscribers based on their topic subscriptions. The captured real-time are displayed with the help of applications as displayed in Fig. 15.

Integrated smart water meter and smart energy meter are implemented in two houses. In House 1, the smart water meter is connected, allowing the capture and display of total water consumption in liters for billing purposes. Meanwhile, in House 2, the smart water meter is disconnected. Additionally, the electricity consumption in units from both House 1 and House 2 smart energy meters is captured and displayed. The smart distribution transformer parameters, including temperature, current, voltage, power factor, power, and frequency, are also displayed. If the consumed power exceeds the allocated limit, an overload status is indicated, as depicted in Fig. 15(b) for House 2.

The applications not only showcase the data but also provides

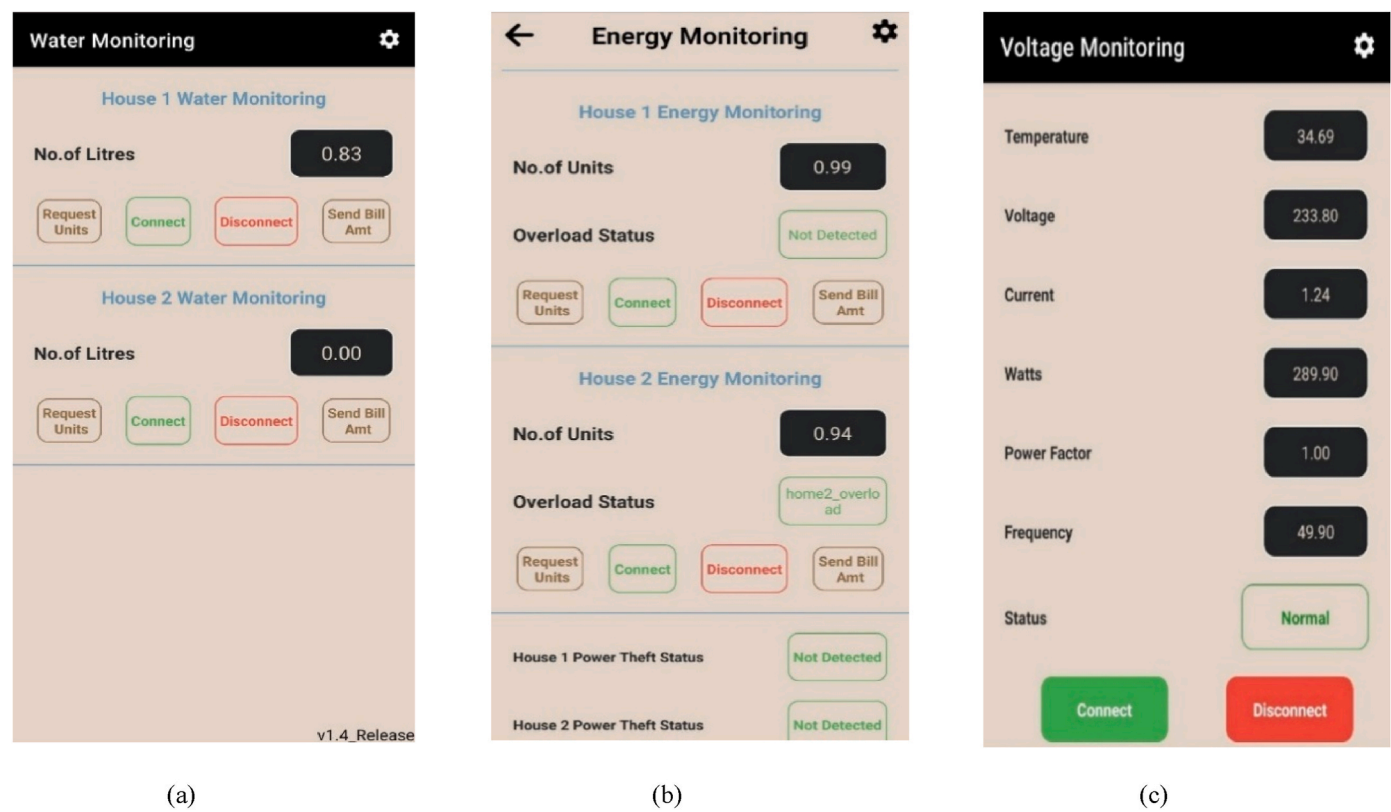


Fig. 15. Data captured by the MQTT applications: (a) Smart water meter data, (b) Smart energy meter data and (c) Smart distribution transformer readings.

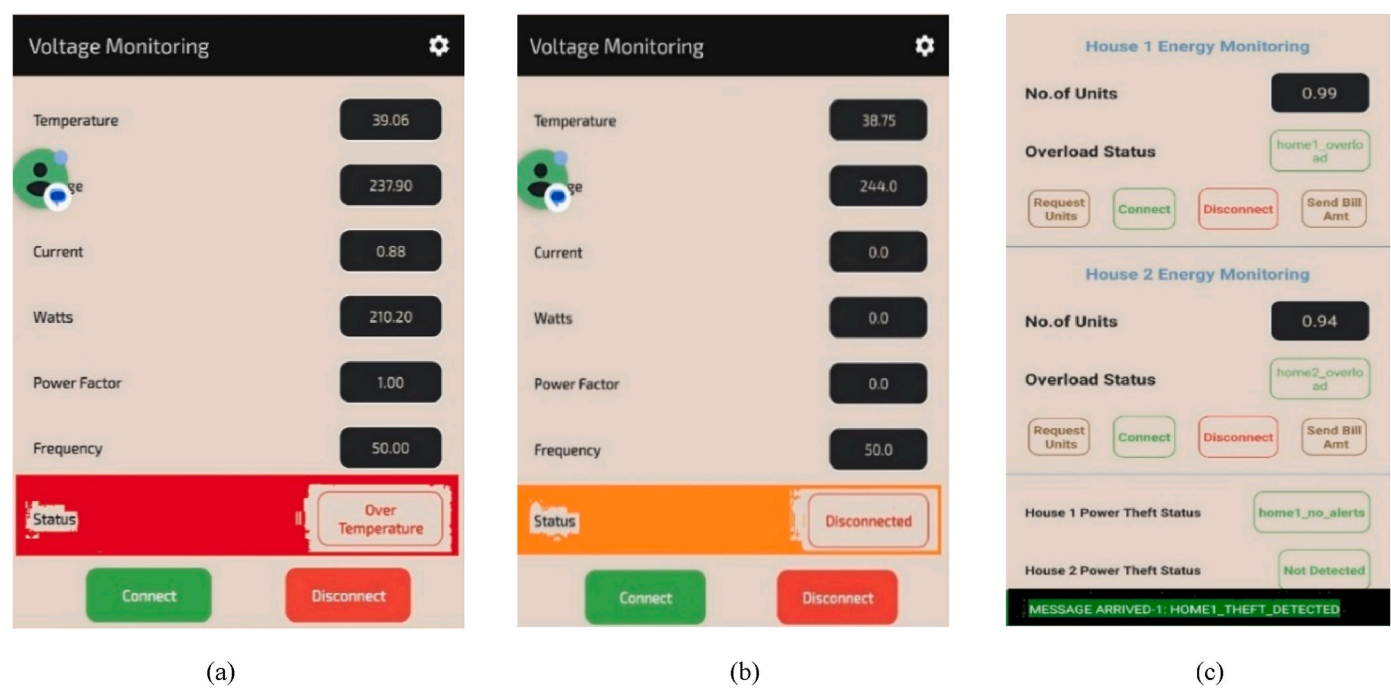
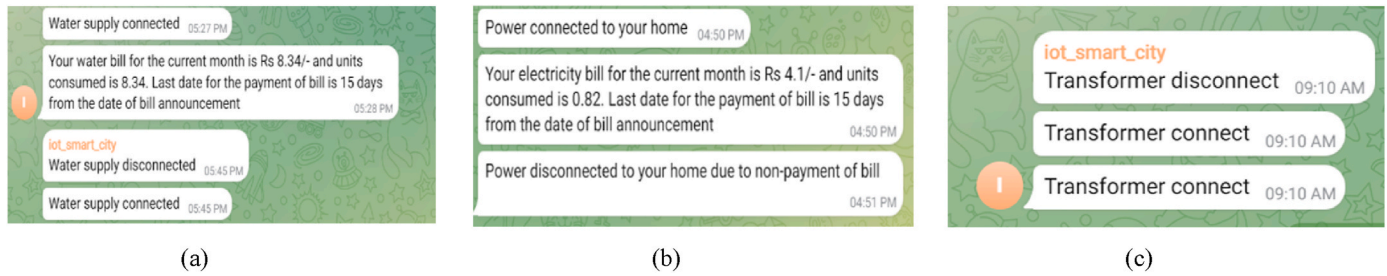


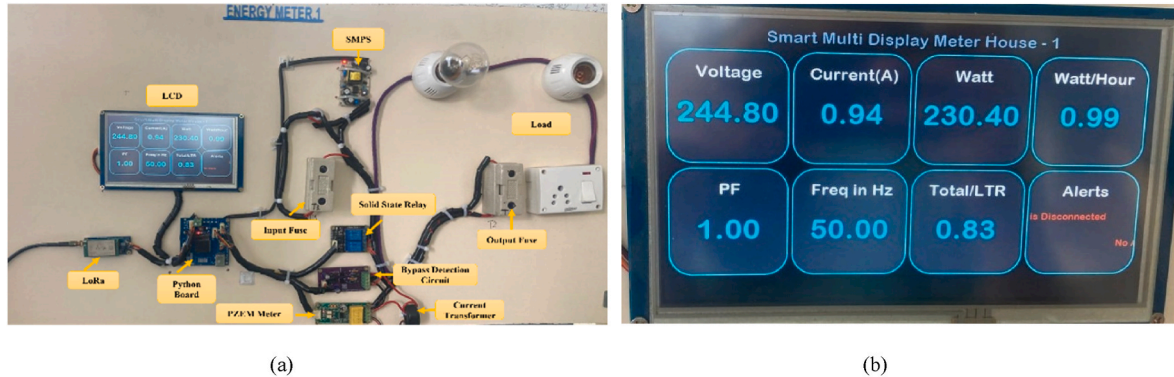
Fig. 16. Data captured by the MQTT applications: (a) Temperature rise in smart distribution transformer, (b) Disconnection of smart distribution transformer due to temperature rise and (c) Electricity theft detection in house 1.

notifications in the event of a malfunction as shown in Fig. 16. A threshold temperature value of 39 °C is defined, and if the temperature surpasses this limit, an immediate disconnection of the distribution transformer occurs to prevent further damage. Similarly, in the event of an overload, the distribution transformer is promptly disconnected. The

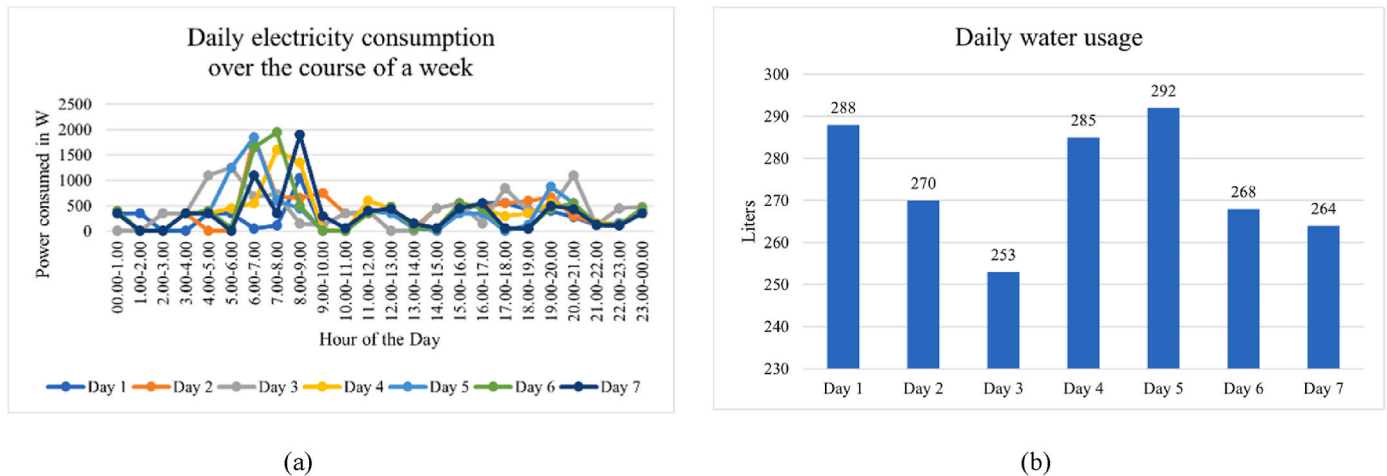
applications capture an overload status for both House 1 and House 2, triggering a buzzer to alert users of the overload. If the overload persists for a predetermined duration, such as 5 min, the billing charges are doubled. Additionally, the applications detect electricity theft in House 1, prompting alerts to authorities for necessary action. The relevant data



**Fig. 17.** Messages communicated through telegram application: (a) Billing information, suspension of water supply due to unpaid bills, and restoration of water supply upon successful bill payment, (b) Restoration of water supply upon successful bill payment, billing information, and suspension of electricity due to unpaid bills and (c) Disconnection and connection alert of distribution transformer.



**Fig. 18.** 7" TFT display at the user side: (a) Configuration of the smart energy meter hardware alongside the TFT display and (b) Presentation of real-time water and energy data on the display, including alerts when applicable.



**Fig. 19.** Power and water consumption: (a) Weekly power usage recorded by a smart energy meter and (b) Weekly water usage monitored by a smart water meter.

is communicated to both the user and regulatory authorities through the Telegram application as shown in Fig. 17.

A 7" Thin Film Transistor (TFT) display is employed to keep consumers informed about their energy and water usage, as well as receive alerts such as water disconnection, electricity disconnection, and power overload. To simplify comprehension, the cells are structured to present each essential data category distinctly. This includes voltage, current, power, units consumed, power factor, frequency, total water consumed, and alerts if any. The Human Machine Interface (HMI) and its GUI for the TFT display is crafted using the Nextion Editor. The TFT display, situated adjacent to the smart energy meter, effectively showcases consumption data in real-time from the integrated water and energy meter. The smart energy meter and smart water meter communicate

seamlessly through the LoRa network, with the results being prominently showcased on the TFT display as seen in Fig. 18. Alert messages associated with the smart energy meter appear in the initial row of the alerts cell, whereas alerts linked to a smart water meter are presented in the second row.

The power consumption of a household for a week, incorporating an LED television, refrigerator of 324-L, an electric water heater, electric fans-03 numbers, a grinder, five LED bulbs of 14-W, and a water purifier, is illustrated in Fig. 19(a). In a similar manner, Fig. 19(b) portrays the water usage of a residence occupied by a three-member family throughout the week.

The three applications effectively retrieved data from the smart water meter, smart energy meter, and smart distribution transformer,

providing real-time information to both users and regulatory bodies. Additionally, all system-related alerts were successfully communicated. The following section delves into a comparative analysis between the current system and the pre-existing one.

#### 4.5. Contrast with existing system

Table 4 outlines the essential distinctions among current water meters, energy meters, and smart distribution transformers, as well as the envisaged integrated smart water and energy meter, and smart distribution transformer. Table 5 outlines the highlights of the envisioned innovative integrated system, along with the smart distribution transformer. The designed system facilitates bidirectional communication, incorporates a smart water meter powered by solar energy, provides remote system control, issues overload warnings, and notifies service providers of energy theft-features that were absent in the current system.

Despite advancements in metering and distribution transformer technology, end-users do not benefit fully. Existing systems lack remote operation, communication with service providers, power theft

detection, and overload and temperature rise warnings. Traditional metering is error-prone, and distribution transformers need frequent on-site checks, risking disasters if neglected. The proposed system aims to address these shortcomings. Existing meters and smart distribution transformers fall into two categories: those that record and store consumption data internally and those that transfer data wirelessly, typically using GSM technology. Retrieving data from memory chips requires manual intervention, leading to delays in billing. Network-based systems may fail if communication is unreliable. Wireless networks like Zigbee and Bluetooth have limited range, while Wi-Fi incurs higher costs for consumers. The paper outlines an innovative system that integrates energy and water meters, measures distribution transformer parameters, and uses LoRaWAN standard for real-time communication, applicable to both outdoor and indoor environments with a 3-mile coverage distance. The system operates independently of cellular networks, allowing remote control of meters and transformers, a feature lacking in current systems. Another highlight is the smart water meter powered by solar energy, contributing to reduced overall energy consumption.

**Table 4**

Comparative analysis of existing smart water meters, smart energy meters and smart distribution transformer.

| Sl. No. | System Designed                                                              | Key Points                                                                                                                  | Demerits                                                                                                                                          |
|---------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Electromagnetic Flow Meter [32]                                              | With an 85 % accuracy rate, the data measured is transmitted to the service provider via GSM, and the system solar powered. | The system falls short in providing remote monitoring, leaving the user uninformed about their daily water consumption.                           |
| 2.      | Water flow meter utilizing Hall Effect sensor technology [33]                | Algorithm designed for predicting and managing water demand.                                                                | No communication with the consumer as well as the service provider.                                                                               |
| 3.      | Piezo Clamps [34]                                                            | Non-invasive approach, it boasts a 96.07 % accuracy and offers detection modes: cold, hot, mixed and no flow.               | Does not gauge water usage.                                                                                                                       |
| 4.      | Arrow Sensor Based Smart Water Meter [35]                                    | Non-contact, high accuracy of 95 %, leakage detection and communication of data using GSM/GPRS.                             | The system falls short in providing remote monitoring.                                                                                            |
| 5.      | Ultrasonic sensor [36]                                                       | Non-intrusive with high precision.                                                                                          | Susceptible to deposition within the surface of the pipe and a deficiency in communication with the utility company.                              |
| 6.      | Laser and IR Sensor [37]                                                     | Touchless, with elevated precision.                                                                                         | Characterized by a substantial cost, increased hardware prerequisites, and a deficiency in communication with both the user and service provider. |
| 7.      | Water Flow Sensor YF-S201 [38]                                               | Detection of water leaks and sending usage data to the user via email.                                                      | Lacks connectivity with service providers, and the use of email remains unaccustomed to a considerable number of individuals.                     |
| 8.      | Water Meter Powered Autonomously [39]                                        | Autonomously powered water meter.                                                                                           | Functions either as a harvester of energy or as meter but not as both simultaneously.                                                             |
| 9.      | Water Meter Powered Autonomously [40]                                        | Generator driven by water turbine.                                                                                          | The Bluetooth transceiver exhibits a limited communication range.                                                                                 |
| 10.     | AT89S52 Microcontroller Based Smart Energy Meter [41]                        | Transmitting information through GSM technology and also displaying it on an LCD screen.                                    | Does not include the functionality to identify electricity theft.                                                                                 |
| 11.     | Smart Energy Meter using PIC Microcontroller [45]                            | Transmitting data through GSM.                                                                                              | Does not possess the capability to identify electricity theft or voltage overload.                                                                |
| 12.     | Smart Energy Meter built on Arduino platform [46]                            | Transmitting data through IoT and Wi-Fi connectivity.                                                                       | Does not include the ability to identify instances of electricity theft or voltage overload.                                                      |
| 13.     | Smart Energy Meter built on Arduino platform [47]                            | Transmission of data via GSM.                                                                                               | Does not possess the capability to identify electricity theft or detect voltage overload.                                                         |
| 14.     | Smart Energy Meter using Arduino [49]                                        | Data transmission through IoT and Wi-Fi.                                                                                    | Does not have the capability to detect voltage overload.                                                                                          |
| 15.     | Smart Energy Meter utilizing Arduino [50]                                    | Data transmission through IoT and Wi-Fi.                                                                                    | Does not incorporate the capability to detect electricity theft or voltage overload.                                                              |
| 16.     | Arduino based Smart Energy Meter [51]                                        | Data communication through GSM, Wi-Fi, and IoT.                                                                             | Does not include the ability to detect electricity theft.                                                                                         |
| 17.     | Energy Meter with Smart Features Using PIC Microcontroller [68]              | Maintains a stable output voltage while continuously monitoring energy consumption.                                         | Lacks communication with the utility company and does not incorporate detection of overload.                                                      |
| 18.     | Energy Meter with Smart Functionality Utilizing IoT and Neural Networks [69] | Both the consumer and service provider receive the consumption data.                                                        | Does not have functionalities such as remote disconnection/reconnection, as well as overload alert.                                               |
| 19.     | Smart Energy Meter utilizing IoT [70]                                        | Monitoring energy consumption in real-time.                                                                                 | Does not have features such as overload alerts, communication with the utility company, and remote disconnection/reconnection of electricity.     |
| 20.     | GSM based Smart Distribution Transformer [53]                                | Communicates on the 433 MHz band with other transformers.                                                                   | Transfer of data over the internet via a GPRS connection may face potential issues.                                                               |
| 21.     | Smart Distribution Transformer utilizing GSM technology [54]                 | Utilizes GSM technology to send SMS alerts to the central control room during emergency situations.                         | Does not have the capability for remote connection and disconnection.                                                                             |
| 22.     | Smart Distribution Transformer based on PIC Microcontroller [55]             | The GPRS module transmitted collected data to an online interface accessible by utility engineers.                          | The system was activated only in emergency situations and lacked real-time monitoring support.                                                    |
| 23.     | Smart Distribution Transformer based on PIC Microcontroller [56]             | Utilizing a ThingSpeak channel to identify any irregularities.                                                              | Lacks an alert feature for abnormalities.                                                                                                         |
| 24.     | Smart Distribution Transformer based on Arduino [57]                         | Utilizing a ThingSpeak channel to identify any irregularities.                                                              | Lacks an alert feature for abnormalities.                                                                                                         |

**Table 5**

Key features of the integrated system proposed.

| Proposed System                                                                         | Key Points                                                                                                                                                                                                                                                                           | Demerits                                                       |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Novel Unified smart water and energy metering system and smart distribution transformer | Simple installation, continuous real-time monitoring, solar-powered water meter, communicates with consumers as well as service providers, remote reconnection and disconnection, overload alerts, temperature rise alerts, power theft detection, and minimized human intervention. | Mandates calibration for alterations in water pipe dimensions. |

## 5. Conclusion

Integration of IoT systems with LoRa based communication access proposed in this article is an attempt to develop sustainable system for real time monitoring of water and energy consumption status. The comprehensive solution outlined in this research introduces features like remote power and water shutoff, disconnection alerts, reconnection of water and power, overloading alerts, and power theft alerts, distinguishing the system from existing alternatives. Additionally, the smart distribution transformer monitors the parameters to prevent potential failures that could result in power outages. This system exhibits a process that reduces the need for manual labor and mitigates the possibility of errors that could be made by humans, resulting in both time and cost savings. The innovative integrated experimental setup is developed for a two-house scenario and has yielded results that record real time information of water and energy consumption as well as the performance of distribution transformer. The smart water meter underwent testing with 1000 ml of water to calibrate its factor to 4.5, achieving an accuracy of 98 %. The smart energy meter effectively identified instances of electricity theft and overload. Furthermore, the smart distribution transformer seamlessly deactivated in response to overvoltage and increased temperature. Additionally, the experimental setup was able to be remotely connect and disconnect.

The platform has contributed to improved sensing and communication, thereby enhancing the adaptability of the system for real-time implementation. Employing LoRa network presents a budget friendly method to ensure the smooth operation of the system. The measured data is transmitted wirelessly through LoRa, and the integrated meter demonstrated its capability for distance communication, reaching a maximum of 3 miles. The innovative unified system effectively oversees the water and energy consumption in a household, while also monitoring the health of the distribution transformer. This makes it well-suited for implementation in smart cities application.

Furthermore, the proposed system facilitates the users to identify as well as resolve any abnormal consumption patterns of energy and water. Therefore, the proposed system distinguishes itself by providing all the essential features a real time monitoring system should possess. This facilitates easy access to data for regulatory boards and consumers, simplifying the supplier process and benefiting both. As a result, it proves to be an innovative and unique solution. Moreover, to enhance the system's effectiveness, a future study will include the implementation and integration of street light system with the existing smart water, energy meters and smart distribution transformer. A central hub will be formulated to gather and store data for subsequent analysis employing Machine Learning algorithms.

## CRedit authorship contribution statement

**Sushma N.:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation,

Conceptualization. **Suresh H.N.:** Writing – review & editing, Supervision, Project administration. **Mohana Lakshmi J.:** Writing – review & editing, Supervision, Project administration. **Santhosh Kumar K. B.:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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